

Abstract

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The **Terraform-Managed Azure Subscription Baseline** project focuses on automating the configuration, governance, and management of an Azure subscription using **Terraform Infrastructure as Code (IaC)**. In traditional cloud environments, administrators often configure resources and security policies manually through the Azure Portal, which can lead to inconsistent configurations, human errors, and difficulty in tracking changes. This project addresses these challenges by defining the required Azure subscription guardrails and configurations as reusable Terraform code.

The primary objective of the project is to establish a **standardized and repeatable Azure subscription baseline** that can be deployed and maintained through code. Terraform enables teams to provision subscription-level configurations consistently and provides a clear record of infrastructure changes. The project also supports **configuration drift detection**, allowing teams to identify differences between the desired infrastructure state defined in Terraform and the actual state of the Azure environment. This is particularly useful when changes are made manually through the Azure Portal.

The project considers important collaboration challenges that may occur when multiple team members work on the same Terraform infrastructure. One major issue is **Terraform state-file locking**, which can occur when multiple contributors attempt to modify the infrastructure simultaneously. Proper state management and controlled collaboration are therefore essential

to prevent conflicts and inconsistent deployments. Another challenge is that manual changes made through the Azure Portal may cause Terraform to generate unexpected or destructive execution plans when the actual infrastructure no longer matches the Terraform configuration.

By managing the Azure subscription through Terraform, the project promotes **automation, consistency, governance, traceability, and reliable infrastructure management**. Overall, the proposed baseline provides a structured approach for maintaining Azure environments while reducing manual intervention and improving the ability of development and operations teams to safely collaborate on cloud infrastructure.